

WOOD FUSION FOR MONOIDAL COMONADS

ROSS STREET

ABSTRACT. The goal is to show how a 1978 paper of Richard Wood on monoidal comonads and exponentiation relates to a more recent publication by Alain Bruguières, Stephen Lack and Alexis Virelizier. In the process, we mildly extend the ideas to the skew-closed setting.

2010 Mathematics Subject Classification: 16E45; 16D90; 18A40

Key words and phrases: Hopf comonad; monoidal comonad; skew-closed category; skew-monoidal category.

CONTENTS

1. Introduction	1
2. Monoidal comonads and exponentials	1
3. Wood fusion morphisms	3
4. Fusion for opmonoidal monads	5
5. Fusion for monoidal comonads	6
References	7

1. INTRODUCTION

This paper began by moving offices and uncovering Richard Wood's paper [24]. At this point I am merely throwing bits and pieces together. They may not yet be said quite right.

2. MONOIDAL COMONADS AND EXPONENTIALS

See [17, 11, 22] for notation and definitions. We work with categories enriched in a nice base symmetric monoidal category $\mathcal{V}$ without always writing the $\mathcal{V}$ prefix.

Let G be a monoidal comonad on a skew-monoidal category $\mathcal{C}$. The comonad structure involves a counit $\varepsilon : G \rightarrow 1_{\mathcal{C}}$ and a comultiplication $\delta : G \rightarrow GG$. The monoidal structure involves a natural family of morphisms $\varphi : GX \otimes GY \rightarrow G(X \otimes Y)$ and a morphism $\varphi_0 : I \rightarrow GI$.

For the non-skew case, the following was noted by Wood in [24] with renewed interest generated by Moerdijk [14]. The category $\mathcal{C}^G$ of Eilenberg-Moore G -coalgebras becomes skew monoidal with tensor product

$$(X \xrightarrow{\xi} GX) \otimes (Y \xrightarrow{v} GY) = (X \otimes Y \xrightarrow{\xi \otimes v} GX \otimes GY \xrightarrow{\varphi} G(X \otimes Y))$$

and skew unit $I \xrightarrow{\varphi_0} GI$.

Date: February 7, 2023.

The author gratefully acknowledges the support of Australian Research Council Discovery DP190102432.

Remark 2.1. If we were dealing with a skew promonoidal category $\mathcal{C}$, with the skew promonoidal tensor

$$P_{\mathcal{C}} : \mathcal{C}^{\text{op}} \otimes \mathcal{C}^{\text{op}} \otimes \mathcal{C} \rightarrow \mathcal{V} ,$$

the skew promonoidal tensor $P_{\mathcal{C}^G}((X, \xi), (Y, v); (Z, \zeta))$ for $\mathcal{C}^G$ would be the equalizer of $P_{\mathcal{C}}(1_X, 1_Y; \zeta)$ and the composite

$$P_{\mathcal{C}}(X, Y; Z) \xrightarrow{\varphi} P_{\mathcal{C}}(GX, GY; GZ) \xrightarrow{P_{\mathcal{C}}(\xi, v; 1_Z)} P_{\mathcal{C}}(X, Y; GZ) .$$

Definition 2.2. For objects X and Y of a skew-monoidal category $\mathcal{C}$, an *exponential of Y by X* is an object $[X, Y]$ of $\mathcal{C}$ equipped with a morphism $e_Y^X : [X, Y] \otimes X \rightarrow Y$, called *evaluation at X* , for which the morphism

$$\mathcal{C}(W, [X, Y]) \xrightarrow{-\otimes X} \mathcal{C}(W \otimes X, [X, Y] \otimes X) \xrightarrow{\mathcal{C}(1, e_Y^X)} \mathcal{C}(W \otimes X, Y)$$

is invertible.

Lemma 2.3. *Let G be a monoidal comonad on a skew-monoidal category $\mathcal{C}$. Let (X, ξ) and (Z, ζ) be objects of $\mathcal{C}^G$. Suppose the exponentials $[(X, \xi), (GZ, \delta_Z)]$ and $[(X, \xi), (G^2Z, \delta_{GZ})]$ exist in $\mathcal{C}^G$. Then the exponential $[(X, \xi), (Z, \zeta)]$ exists if and only if the parallel pair*

$$[(X, \xi), (GZ, \delta_Z)] \begin{array}{c} \xrightarrow{[1, \delta_Z]} \\ \xrightarrow{[1, G\zeta]} \end{array} [(X, \xi), (G^2Z, \delta_{GZ})] . \quad (2.1)$$

has an equalizer $(E, \kappa) \xrightarrow{k} [(X, \xi), (GZ, \delta_Z)]$ in $\mathcal{C}^G$. When that holds, $[(X, \xi), (Z, \zeta)] = (E, \kappa)$ and the evaluation $e_{(Z, \zeta)}^{(X, \xi)}$ is determined by commutativity of the square (2.2).

$$\begin{array}{ccc} (E, \kappa) \otimes (X, \xi) & \xrightarrow{e_{(Z, \zeta)}^{(X, \xi)}} & (Z, \zeta) \\ k \otimes 1 \downarrow & & \downarrow \zeta \\ [(X, \xi), (GZ, \delta_Z)] \otimes (X, \xi) & \xrightarrow{e_{(GZ, \delta_Z)}^{(X, \xi)}} & (GZ, \delta_Z) \end{array} \quad (2.2)$$

Proof. We have the equalizer

$$(Z, \zeta) \xrightarrow{\zeta} (GZ, \delta_Z) \begin{array}{c} \xrightarrow{\delta_Z} \\ \xrightarrow{G\zeta} \end{array} (G^2Z, \delta_{GZ}) \quad (2.3)$$

in $\mathcal{C}^G$. The “only if” direction is then clear since right adjointness of the functor $[(X, \xi), -]$ implies it preserves the equalizer (2.3). For the other direction, note that morphisms $f : (A, \alpha) \rightarrow (E, \kappa)$ are in bijection (via composition with k) with morphisms $g : (A, \alpha) \rightarrow [(X, \xi), (GZ, \delta_Z)]$ such that $[1, \delta_Z] \circ g = [1, G\zeta] \circ g$ and so with morphisms $h : (A, \alpha) \otimes (X, \xi) \rightarrow (GZ, \delta_Z)$ such that $\delta_Z \circ h = G\zeta \circ h$. Using the equalizer (2.3), we see that (E, κ) is the exponential and that the last sentence of the lemma holds. $\square$

The following lemma is essentially in [24].

Lemma 2.4. *Let G be a monoidal comonad on a skew-monoidal category $\mathcal{C}$. Suppose the exponentials $[X, Y]$ and $[GX, GY]$ exist in $\mathcal{C}$. Then an exponential of the cofree object (GY, δ_Y) by any object (X, ξ) in $\mathcal{C}^G$ is given by the cofree object $(G[X, Y], \delta_{[X, Y]})$ with the composite*

$$G[X, Y] \otimes X \xrightarrow{\varphi \otimes \xi} [GX, GY] \otimes GX \xrightarrow{e_{GY}^{GX}} GY$$

as evaluation.

Proof. We have the chain of isomorphisms

$$\begin{aligned}
& \mathcal{C}^G((A, \alpha), (G[X, Y], \delta_{[X, Y]})) \\
& \cong \mathcal{C}(A, [X, Y]) \\
& \cong \mathcal{C}(A \otimes X, Y) \\
& \cong \mathcal{C}^G((A \otimes X, \varphi(\alpha \otimes \xi)), (GY, \delta_Y)) \\
& \cong \mathcal{C}^G((A, \alpha) \otimes (X, \xi), (GY, \delta_Y))
\end{aligned}$$

which by Yoneda is induced by the evaluation morphism given by the clockwise path around the diagram (2.4).

$$\begin{array}{ccccc}
G[X, Y] \otimes X & \xrightarrow{\delta_{[X, Y]} \otimes \xi} & GG[X, Y] \otimes GX & \xrightarrow{\varphi} & G(G[X, Y] \otimes X) \\
& \searrow 1 \otimes \xi & \downarrow G\varepsilon_{[X, Y]} \otimes 1 & & \downarrow G(\varepsilon_{[X, Y]} \otimes 1) \\
& & G[X, Y] \otimes GX & \xrightarrow{\varphi} & G([X, Y] \otimes X) \\
& \searrow \varphi \otimes \xi & \downarrow \hat{\varphi} \otimes 1 & & \downarrow Ge_Y^X \\
& & [GX, GY] \otimes GX & \xrightarrow{e_{GY}^{GX}} & GY
\end{array} \tag{2.4}$$

Diagram (2.4) thus shows the evaluation is equal to the composite stated in the lemma. $\square$

3. WOOD FUSION MORPHISMS

Notice that, if the exponential $[X, GY]$ exists, then the evaluation in Lemma 2.4 corresponds, under the universal property of exponential, to the composite

$$w_{\xi, Y} := G[X, Y] \xrightarrow{\hat{\varphi}} [GX, GY] \xrightarrow{[\xi, 1]} [X, GY]. \tag{3.5}$$

Definition 3.1. The $w_{\xi, Y}$ of (3.5) are called *Wood G -fusion morphisms*.

Notice that the Wood fusion morphism for cofree coalgebras occurs in the construction of a new skew-closed structure using a closed comonad; see Proposition 3 of [22].

We will be interested in when the Wood fusion morphisms are invertible. As with ordinary fusion (recalled in Section 4), invertibility for an arbitrary G -coalgebra follows from invertibility for cofree G -coalgebras.

Proposition 3.2. *For a G -coalgebra (X, ξ) and any object Y , the Wood fusion morphism $w_{\xi, Y}$ is invertible if $w_{\delta_X, Y}$ is invertible and $w_{\delta_{GX}, Y}$ is an epimorphism.*

Proof. For any coalgebra (Z, ζ) , we have an equalizer of the form

$$\begin{array}{ccccc}
Z & \xrightarrow{\zeta} & GZ & \begin{array}{c} \xrightarrow{\delta_Z} \\ \xrightarrow{G\zeta} \end{array} & G^2Z
\end{array} \tag{3.6}$$

which is preserved by all functors (that is, it is an absolute equalizer). It follows that the rows of (3.7) are coequalizers. The vertical morphisms give two composable morphisms of coequalizer diagrams (that is, the appropriate diagrams commute using naturality of ψ , and

coassociativity properties of γ and δ).

$$\begin{array}{ccccc}
G[G^2X, Y] & \xrightarrow{G[\delta_A, 1]} & G[GX, Y] & \xrightarrow{G[\gamma, 1]} & G[X, Y] \\
\downarrow \hat{\varphi} & \xrightarrow{[G\delta_A, 1]} & \downarrow \hat{\varphi} & \xrightarrow{[G\gamma, 1]} & \downarrow \hat{\varphi} \\
[G^3X, GY] & \xrightarrow{[G^2\gamma, 1]} & [G^2X, GY] & \xrightarrow{[G\gamma, 1]} & [GX, GY] \\
\downarrow [\delta_{GA}, 1] & \xrightarrow{[\delta_A, 1]} & \downarrow [\delta_A, 1] & \xrightarrow{[\gamma, 1]} & \downarrow [\gamma, 1] \\
[G^2X, GY] & \xrightarrow{[G\gamma, 1]} & [GX, GY] & \xrightarrow{[\gamma, 1]} & [X, GY]
\end{array} \tag{3.7}$$

If the middle vertical composite is invertible and the left vertical composite is an epimorphism then the right vertical composite is invertible. $\square$

Definition 3.3. The closed comonad G on $\mathcal{C}$ is *Hopf-Wood* when the Wood fusion morphisms $w_{\xi, Y}$ are all invertible. By Proposition 3.2, it suffices to know that all $w_{\delta_X, Y}$ are invertible; then the property can be expressed without reference to Eilenberg-Moore coalgebras.

Lemma 3.4. *Under the conditions of Lemma 2.4, the parallel pair (2.1) is isomorphic to the parallel pair*

$$(G[X, Z], \delta_{[X, Z]}) \xrightarrow[G[1, \zeta]]{Gw_{\xi, Z} \circ \delta_{[X, Z]}} (G[X, GZ], \delta_{[X, GZ]}) . \tag{3.8}$$

A strong monoidal functor $K : \mathcal{A} \rightarrow \mathcal{C}$ is said to *create the exponential of B by A* in $\mathcal{A}$ when the exponential of KB by KA exists in $\mathcal{C}$, and there exist H in $\mathcal{A}$ and $\tau : KH \cong [KA, KB]$ in $\mathcal{C}$ with the following two properties:

- (i) there exists a unique morphism $\bar{e} : H \otimes A \rightarrow B$ such that the square

$$\begin{array}{ccc}
KH \otimes KA & \xrightarrow{\tau \otimes 1} & [KA, KB] \otimes KA \\
\downarrow \varphi & & \downarrow e_{KB}^{KA} \\
K(H \otimes A) & \xrightarrow{K\bar{e}} & KB
\end{array} \tag{3.9}$$

commutes;

- (ii) the object H with $\bar{e}$ is an exponential for B by A .

Lemma 3.5. *Suppose $[X, Y]$ and $[GX, GY]$ exist in $\mathcal{C}$. The forgetful functor $U_G : \mathcal{C}^G \rightarrow \mathcal{C}$ creates the exponential $[(X, \xi), (GY, \delta_Y)]$ of Lemma 2.4 if and only if $[X, GY]$ exists in $\mathcal{C}$ and the Wood G -fusion morphism (3.5) is invertible.*

Proof. Assume $[X, GY]$ exists in $\mathcal{C}$ and the Wood G -fusion morphism (3.5) is invertible. In the definition of creation, take $H = (G[X, Y], \delta_{[X, Y]})$ and $\tau = w_{\xi, Y}$. Commutativity of diagram (3.9) means, in this case, that $\bar{e}$ must be the clockwise route around the diagram

$$\begin{array}{ccccc}
G[X, Y] \otimes X & \xrightarrow{\hat{\varphi} \otimes 1} & [GX, GY] \otimes X & \xrightarrow{[\xi, 1] \otimes 1} & [X, GY] \otimes X \\
\downarrow 1 \otimes \xi & & \downarrow 1 \otimes \xi & & \downarrow e_{GY}^X \\
G[X, Y] \otimes GX & \xrightarrow{\hat{\varphi} \otimes 1} & [GX, GY] \otimes GX & \xrightarrow{e_{GY}^{GX}} & GY
\end{array}$$

and the diagram shows that $\bar{e}$ is the evaluation displayed in Lemma 2.4. By Lemma 2.4 we have what we need for “if”.

Now suppose $U_G : \mathcal{C}^G \rightarrow \mathcal{C}$ creates the exponential $[(X, \xi), (GY, \delta_Y)]$ which we know from Lemma 2.4 is the object $(G[X, Y], \delta_{[X, Y]})$ with the evaluation $\bar{e}$ as displayed in that

lemma. Since $U_G(GY, \delta_Y) = GY$ and $U_G(X, \xi) = X$, we have the existence of $[X, GY]$ and that there is an isomorphism $\tau : G[X, Y] \cong [X, GY]$ such that $\bar{e} = e_{GY}^X \circ (\tau \otimes 1)$. This last equation means that τ corresponds to $\bar{e}$ under the universal property of $[X, GY]$; that is, $\tau = w_{\xi, Y}$, which proves “only if”. $\square$

Proposition 3.6. *Let G be a monoidal comonad on a skew-monoidal category $\mathcal{C}$. Suppose $(X, \xi) \in \mathcal{C}^G$ is such that $[X, Y]$ and $[GX, GY]$ exist for all $Y \in \mathcal{C}$. The Wood G -fusion morphisms $w_{\xi, Y}$ are invertible for all Y if and only if $U_G : \mathcal{C}^G \rightarrow \mathcal{C}$ creates all exponential by (X, ξ) . In this case, for any $(Z, \zeta) \in \mathcal{C}^G$,*

$$[(X, \xi), (Z, \zeta)] \cong ([X, Z], w_{\xi, Z}^{-1} \circ [1, \zeta]) .$$

Proof. Lemma 3.5 gives “if”. Suppose all $w_{(X, \xi), Y}$ are invertible. We will use facts involved in the Beck monadicity theorem [12] in dual form for comonads. We have the cosplit equalizer (3.6). From Lemma 3.5, the parallel pair in equalizer (2.1) is taken by U_G to the cosplit pair

$$[X, GZ] \begin{array}{c} \xrightarrow{[1, \delta_Z]} \\ \xrightarrow{[1, G\zeta]} \end{array} [X, G^2 Z] \quad (3.10)$$

which, by applying $[X, -]$ to (3.6), has the cosplit equalizer

$$[X, Z] \xrightarrow{[1, \zeta]} [X, GZ] \begin{array}{c} \xrightarrow{[1, \delta_Z]} \\ \xrightarrow{[1, G\zeta]} \end{array} [X, G^2 Z] . \quad (3.11)$$

Since U_G is comonadic, there exists a unique G -coalgebra $([X, Z], \kappa)$ and an equalizer

$$([X, Z], \kappa) \xrightarrow{[1, \zeta]} ([X, GZ], \kappa_1) \begin{array}{c} \xrightarrow{[1, \delta_Z]} \\ \xrightarrow{[1, G\zeta]} \end{array} ([X, G^2 Z], \kappa_2) \quad (3.12)$$

in $\mathcal{C}^G$, where (using Lemma 3.4) the coactions κ_1 and κ_2 are transported from the coactions $\delta_{[X, Z]}$ on $G[X, Z]$ and $\delta_{[X, GZ]}$ on $G[X, GZ]$ under the invertible Wood fusion morphisms. So we have condition (ii) for U_G to create the exponentiation. For condition (i), note that commutativity of (2.2) in Lemma 2.3 with $k = [1, \zeta]$ shows that evaluation $e_Z^X : ([X, Z], \kappa) \otimes (X, \xi) \rightarrow (Z, \zeta)$ is the G -coalgebra morphism for the unique solution to diagram (3.9). $\square$

4. FUSION FOR OPMONOIDAL MONADS

Let T be an opmonoidal monad on the skew-monoidal category $\mathcal{C}$. The monad structure involves a unit $\eta : 1_{\mathcal{C}} \rightarrow T$ and a multiplication $\mu : TT \rightarrow T$. The opmonoidal structure involves a natural family of morphisms $\psi : T(X \otimes Y) \rightarrow TX \otimes TY$ and a morphism $\psi_0 : TI \rightarrow I$. We denote the skew-monoidal category of Eilenberg-Moore T -algebras by $\mathcal{C}^T$ with strong monoidal forgetful functor $U_T : \mathcal{C}^T \rightarrow \mathcal{C}$. The tensor product for $\mathcal{C}^T$ is defined by

$$(TX \xrightarrow{\alpha} X) \otimes (TY \xrightarrow{\beta} Y) = (T(X \otimes Y) \xrightarrow{\psi} TX \otimes TY \xrightarrow{\alpha \otimes \beta} X \otimes Y)$$

and skew unit $TI \xrightarrow{\psi_0} I$.

For $W \in \mathcal{C}$ and $(X, \alpha) \in \mathcal{C}^T$, we call the composite $v = v_{W, \alpha}$:

$$T(W \otimes X) \xrightarrow{\psi} TW \otimes TX \xrightarrow{1 \otimes \alpha} TW \otimes X \quad (4.13)$$

a T -fusion morphism (as featured in [3]).

Suppose $T : \mathcal{C} \rightarrow \mathcal{C}$ has a right adjoint functor G . As discussed in [8], G becomes a comonad on $\mathcal{C}$ and there is an isomorphism of categories $\mathcal{C}^T \cong \mathcal{C}^G$ over $\mathcal{C}$. These matters

involve the calculus of mates (in the sense of [9]) as does the fact that G becomes a monoidal comonad and the isomorphism $\mathcal{C}^T \cong \mathcal{C}^G$ becomes strong monoidal.

Proposition 4.1. *Suppose T is an opmonoidal monad on the skew-monoidal category $\mathcal{C}$. Suppose G is a right adjoint monoidal comonad for T . Let $(X, \alpha) \in \mathcal{C}^T$ correspond to $(X, \xi) \in \mathcal{C}^G$. The T -fusion morphism $v_{W, \alpha}$ is invertible for all $W \in \mathcal{C}$ if and only if the Wood G -fusion morphism $w_{\xi, Y}$ is invertible for all $Y \in \mathcal{C}$.*

Proof. Apply the Yoneda Lemma to the following commutative diagram where $\sigma : TG \rightarrow 1_{\mathcal{C}}$ is the counit of $T \dashv G$.

$$\begin{array}{ccc}
 \mathcal{C}(TW \otimes X, Y) & \xrightarrow{\cong} & \mathcal{C}(W, G[X, Y]) \\
 \downarrow \mathcal{C}(1 \otimes \alpha, 1) & \searrow \mathcal{C}(1 \otimes \sigma_X, 1) & \downarrow \mathcal{C}(1, \varphi) \\
 & \mathcal{C}(TW \otimes TGX, Y) & \\
 \downarrow \mathcal{C}(1 \otimes T\xi, 1) & \swarrow \mathcal{C}(\psi, 1) & \downarrow \mathcal{C}(1, [\varphi]) \\
 \mathcal{C}(TW \otimes TX, Y) & & \mathcal{C}(W, [GX, GY]) \\
 \downarrow \mathcal{C}(\psi, 1) & \searrow \mathcal{C}(T(1 \otimes \xi), 1) & \downarrow \mathcal{C}(1, [\xi, 1]) \\
 & \mathcal{C}(T(W \otimes GX), Y) & \\
 \downarrow \mathcal{C}(T(1 \otimes \xi), 1) & \swarrow \mathcal{C}(T(1 \otimes \xi), 1) & \downarrow \mathcal{C}(1, [\xi, 1]) \\
 \mathcal{C}(T(W \otimes X), Y) & \xrightarrow{\cong} & \mathcal{C}(W, [X, GY])
 \end{array}$$

where $\sigma : TG \rightarrow 1_{\mathcal{C}}$ is the counit of $T \dashv G$. □

Example 4.2. Let H be a bimonoid in a braided closed monoidal category $\mathcal{C}$. Then $- \otimes H : \mathcal{C} \rightarrow \mathcal{C}$ is an opmonoidal monad with right adjoint $[H, -] : \mathcal{C} \rightarrow \mathcal{C}$. Proposition 4.1 relates Wood fusion for the monoidal comonad $[H, -]$ with the fusion morphisms for the bimonoid H . All are invertible if and only if H is Hopf. In this case $- \otimes H$ is Hopf and $[H, -]$ is Hopf-Wood.

5. FUSION FOR MONOIDAL COMONADS

A monoidal comonad G on a skew-monoidal category $\mathcal{C}$ can be considered to be an opmonoidal monad G^{op} on the skew-monoidal category $\mathcal{C}^{\text{op}}$. With $T = G^{\text{op}}$, the T -fusion morphism (4.13) becomes the composite $v = v_{W, \xi}$:

$$GW \otimes X \xrightarrow{1 \otimes \xi} GW \otimes GX \xrightarrow{\varphi} G(W \otimes X) \quad (5.14)$$

which is a G -fusion morphism (as featured in [4]) for $W \in \mathcal{C}$ and $(X, \xi) \in \mathcal{C}^G$.

Consider the Yoneda embedding

$$\mathfrak{y} : \mathcal{C}^{\text{op}} \longrightarrow [\mathcal{C}, \mathcal{V}] \quad (5.15)$$

taking X to the representable functor $\mathcal{C}(X, -)$. The monoidal structure on $[\mathcal{C}, \mathcal{V}]$ is Day convolution $M \star N$ and gives the monoidal cocompletion of $\mathcal{C}^{\text{op}}$. The opmonoidal monad G^{op} on $\mathcal{C}^{\text{op}}$ extends to a left adjoint opmonoidal monad Lan_G on $[\mathcal{C}, \mathcal{V}]$ defined by left Kan extension along $G : \mathcal{C} \rightarrow \mathcal{C}$. Each G -coalgebra (X, ξ) gives a Lan_G -algebra $(\mathfrak{y}X, \mathfrak{y}\xi)$ since we can identify $\text{Lan}_G \mathcal{C}(X, -)$ with $\mathcal{C}(GX, -)$. Write Res_G for the right adjoint monoidal comonad to Lan_G given by restriction $[G, 1_{\mathcal{C}}]$ along G .

Proposition 5.1. *In the situation just described, the following conditions are equivalent:*

- (i) *the G -fusion morphism $v = v_{W, \xi}$ is invertible for all $W \in \mathcal{C}$;*

- (ii) the Lan_G -fusion morphism $v = v_{M, \star \xi}$ is invertible for all $M \in [\mathcal{C}, \mathcal{V}]$;
- (iii) the Wood Res_G -fusion morphism $w = w_{\star \xi, N}$ is invertible for all $N \in [\mathcal{C}, \mathcal{V}]$.

Proof. The equivalence of (ii) and (iii) follows from Proposition 4.1. Using some coend manipulation, or otherwise, we see that the composite

$$\text{Lan}_G(M \star \star X) \xrightarrow{\psi} \text{Lan}_G M \star \text{Lan}_G \star X \xrightarrow{1 \star \star \xi} \text{Lan}_G M \star \star X$$

is isomorphic to the composite obtained by applying $\int^W - \otimes MW$ to

$$\mathcal{C}(G(W \otimes X), -) \xrightarrow{\mathcal{C}(\varphi, 1)} \mathcal{C}(GW \otimes GX, -) \xrightarrow{\mathcal{C}(1 \otimes \xi, 1)} \mathcal{C}(GW \otimes X, -).$$

This shows the equivalence of (ii) and (i). $\square$

Write $\llbracket X, Y \rrbracket$ for the exponential of Y by X in $\mathcal{C}^{\text{op}}$. It is the *co-exponential* of Y by X in $\mathcal{C}$ and exists if and only the exponential of $\star Y$ by $\star X$ is in the essential image of (5.15). Now we can recover results of [3, 4, 16] by applying Proposition 5.1 to the pullback (5.16).

$$\begin{array}{ccc} (\mathcal{C}^G)^{\text{op}} & \xrightarrow{\quad} & [\mathcal{C}, \mathcal{V}]^{\text{Res}_G} \\ U_G^{\text{op}} \downarrow & & \downarrow U_{\text{Res}_G} \\ \mathcal{C}^{\text{op}} & \xrightarrow{\quad \star \quad} & [\mathcal{C}, \mathcal{V}] \end{array} \quad (5.16)$$

Corollary 5.2. *Suppose G is a monoidal comonad on the skew-monoidal category $\mathcal{C}$ and suppose (X, ξ) is a G -coalgebra. The G -fusion morphism $v = v_{W, \xi}$ is invertible for all $W \in \mathcal{C}$ if and only if $U_G : \mathcal{C}^G \rightarrow \mathcal{C}$ creates all co-exponentials $\llbracket (X, \xi), (Z, \zeta) \rrbracket$ by (X, ξ) .*

REFERENCES

- [1] BARR, M. AND WELLS, C. *Toposes, Triples and Theories*. Grundlehren der math. Wissenschaften 278, Springer-Verlag, 1985.
- [2] BECK, J.M. Triples, algebras and cohomology. *Reprints in Theory and Applications of Categories* 2 (2003) 1–59.
- [3] BRUGUIÉRES, A., LACK, S. AND VIRELIZIER, A. Hopf monads on monoidal categories. *Advances in Mathematics* 227(2) (2011) 745–800. [5](#), [7](#)
- [4] CHIKHLADZE, D., LACK, S. AND STREET, R. Hopf monoidal comonads. *Theory and Applications of Categories* 24(19) (2010) 554–563. [6](#), [7](#)
- [5] DAY, B.J. Biclosed bicategories: Localisation of convolution. *Macquarie Mathematics Report 81-0030* (Macquarie University, April 1981). <https://arxiv.org/abs/0705.3485>
- [6] DAY, B.J. AND STREET, R. Monoidal bicategories and Hopf algebroids. *Advances in Math.* 129 (1997) 99–157.
- [7] EILENBERG, S. AND KELLY, G.M. Closed categories. *Proceedings of the Conference on Categorical Algebra (La Jolla, 1965)* (Springer-Verlag, 1966) 421–562.
- [8] EILENBERG, S. AND MOORE, J.C. Adjoint functors and triples. *Illinois Journal of Mathematics* 9 (1965) 381–398. [5](#)
- [9] KELLY, G.M. AND STREET, R. Review of the elements of 2-categories. *Lecture Notes in Mathematics* 420 (Springer-Verlag, 1974) 75–103. [6](#)
- [10] LACK, S. Icons. *Applied Categorical Structures* 18(3) (2010) 289–307.
- [11] LACK, S. AND STREET, R. Skew monoidals, skew warpings and quantum categories. *Theory and Applications of Categories* 26(15) (2012) 385–402. [1](#)
- [12] MAC LANE, S. *Categories for the Working Mathematician*. Graduate Texts in Mathematics 5 (Springer-Verlag, 1971). [5](#)
- [13] MCCRUDDEN, P. Opmonoidal monads. *Theory and Applications of Categories* 10 (2002) 469–485.
- [14] MOERDIJK, I. Monads on tensor categories. *Journal of Pure Appl. Algebra* 168 (2002) 189–208.

- [15] NIKOLIĆ, B. *Morphisms of 2-dimensional structures with applications*. Macquarie University, PhD (2018).
<https://protect-au.mimecast.com/s/ac6QCWLvn6ipjvyVt6mNKs?domain=hdl.handle.net>
- [16] NIKOLIĆ, B. AND STREET, R. Comonadic base change for enriched categories. *J. Pure Appl. Algebra (to appear)*.
<http://arxiv.org/abs/1809.02356>. 7
- [17] STREET, R. The formal theory of monads. *Journal of Pure and Applied Algebra* 2(2) (1972) 149–168. 1
- [18] STREET, R. Enriched categories and cohomology with author commentary. *Reprints in Theory and Applications of Categories* 14 (2005) 1–18.
- [19] STREET, R. Absolute colimits in enriched categories. *Cahiers de topologie et géométrie différentielle* 24 (1983) 377–379.
- [20] STREET, R. Fusion operators and cocycloids in monoidal categories. *Applied Categorical Structures* 6 (2) (1998) 177–191.
- [21] STREET, R. *Quantum Groups: A Path to Current Algebra*. Australian Mathematical Society Lecture Series 19. Cambridge University Press, 2007.
- [22] STREET, R. Skew-closed categories. *Journal of Pure and Applied Algebra* 217(6) (June 2013) 973–988. 1, 3
- [23] WEBER, M. Algebraic Kan extensions along morphisms of internal algebra classifiers. *Tbilisi Math. J.* 9(1) (2016) 65–142.
- [24] WOOD, R.J. Coalgebras for closed comonads. *Communications in Algebra* 6(14) (1978) 1497–1504. 1, 2

SCHOOL OF MATHEMATICAL AND PHYSICAL SCIENCES, MACQUARIE UNIVERSITY, NSW 2109, AUSTRALIA

E-mail address: `ross.street@mq.edu.au`